ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech. (CSE)/B.Tech. MBA

Semester: VIII

SUBJECT CODE: BCS091

CRYPTOGRAPHY AND NETWORK SECURITY

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1. Define the following questions: (1 × 5 = 5 Marks)

(i) Define cryptography. Explain its two main types.

(ii) What is the difference between symmetric and asymmetric key cryptography?

(iii) Describe the function of a hash function in cryptography.

(iv) What is the purpose of a Certificate Authority (CA) in Public Key Infrastructure (PKI)?

(v) Define a firewall. What are its types?

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) Define non-repudiation.

(ii) Differentiate between stream cipher and block cipher.

(iii) What is SSL and why is it important in network security?

(iv) What is an example of a block cipher?

(v) List out any four active attacks

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) Encrypt the message "This is an Exercise" with the keyword "Dollars" with the steps involved in Playfair cipher.

OR

(b) Explain Eular's totient function and explain how it is Computed with a suitable example.

(ii)(a) What do you mean by Chiness Remainder Theorem? Solve the following congruent equations by Chiness Remainder Theorem.

$X \equiv 2 \bmod 3$

$X \equiv 1 \bmod 4$

$X \equiv 3 \bmod 5$

OR

(b) What do you understand by Fiestal Cipher Structure? Explain with suitable block diagram.

Q.4.(i) (a) Describe various schemes used for public key distribution.

OR

(b) Discuss the functionality of S/MIME.

(ii)(a) Find the GCD of 252 and 105 using Euclid's algorithm.

OR

(b) Explain and prove the Extended Euclidian Algorithm for finding Multiplicative inverse with a suitable example.

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5.(a). Given:

- A large prime number p=23.
- A primitive root (base) g=5.

Both Alice and Bob will exchange keys. Their private keys are:

- Alice's private key a=6.
- Bob's private key b=15.

Find: The shared secret key.

OR

(b) Explain about the architecture and certification for Kerberos environment. Also differentiate between Kerberos version 4.0 and version 5.0.

Q.6.(a) Discuss the basic usage of message authentication code (MAC) with all the suitable diagram.

OR

(b) Draw and explain the block diagram of DES. Also explain any two block cipher modes of operations with a suitable diagram.